

# XIANG LI

✉ [lx10077@upenn.edu](mailto:lx10077@upenn.edu)    ☎ (+1) 445-544-0286    🌐 [Homepage](#)    🎓 [Google Scholar](#)    🐙 [GitHub](#)

## EDUCATION

---

**Peking University, Beijing, China** Sep. 2018 – Jul. 2023  
Ph.D. in Statistics, School of Mathematical Sciences  
Advisor: [Prof. Zhihua Zhang](#)

**Peking University, Beijing, China** Sep. 2014 – Jul. 2018  
B.S. in Statistics, School of Mathematical Sciences  
B.A. in Economics, National School of Development

## EXPERIENCE

---

**University of Pennsylvania, Philadelphia, USA** Aug. 2023 – Present  
Postdoctoral Researcher, Department of Statistics and Data Science  
Working with [Prof. Weijie Su](#) and [Prof. Qi Long](#)

**University of Toronto, Toronto, Canada** Feb. 2022 – Feb. 2023  
Visiting Ph.D. Student, Department of Statistical Sciences  
Working with [Prof. Qiang Sun](#)

**Face++ (Megvii), Beijing, China** Jun. 2018 – Nov. 2018  
Research Intern, Algorithm Group  
Mentor: [Dr. Shuchang Zhou](#)

**ToSimple, Beijing, China** Mar. 2018 – May. 2018  
Research Intern, Algorithm Group  
Mentor: Dr. Naiyan Wang

## SELECTED AWARDS

---

- IMS New Researcher Travel Award 2025
- Outstanding PhD Graduates, Peking University 2023
- President Scholarship for Excellent Ph.D Student, Peking University 2018, 2020-2021
- Outstanding reviewer in NeurIPS, top 8% 2021
- National Scholarship, China 2017, 2019
- Travel Award NeurIPS 2019, AAAI 2020
- Outstanding Graduates, School of Mathematical Sciences, Peking University 2018
- First prize in National Undergraduate Mathematical Modeling Contest, China 2016

## RESEARCH INTERESTS

---

My research lies at the intersection of statistics, optimization, and machine learning. Currently, I focus on principled and scalable methodologies for generative AI—particularly large language models—with the goal of advancing theoretical understanding and developing insights that support safe, robust, and reliable deployment. Specifically, my research spans the following directions:

- **Statistical Foundations of Generative AI**

- **Provenance and Safety Mechanisms.** Designing watermarking schemes [J4, S2, S3] and developing detection [J1, J2, C3, W1] and estimation [J3] methods for tracing and verifying AI outputs, with the broader aim of building scalable safety mechanisms that safeguard provenance, accountability, and responsible deployment.
- **Fundamental Principles.** Modeling generative AI systems as black boxes and developing statistical tools to analyze input–output relationships, focusing on core issues such as evaluation [S1], uncertainty quantification [S4], scaling law, privacy preserving, and model collapse.
- **Reliable Statistical Machine Learning**
  - **Learning under Challenging Conditions.** Developing efficient algorithms resilient to nonstandard conditions such as heavy-tailed noise [J8], corruption [C5], high-dimensionality [S5], and data heterogeneity, and analyzing their statistical or convergence properties [C12, C7].
  - **Uncertainty Quantification in Adaptive Environments.** Designing statistical methods for streaming data [J5, S7] and adaptive decision-making [C6, C11, C13], ensuring valid statistical inference in dynamic or intricate environments [C2, S8, J6, S6].
- **Towards Efficient Federated Learning**
  - **Communication Efficiency.** Developing communication-efficient algorithms [J9, C10, T2] and analyzing their convergence properties under heterogeneous data settings [C1, C8, T1].
  - **Trustworthiness.** Exploring key aspects of trustworthy federated learning, including differential privacy [C4, S9] and personalization [J7, C9].

## PUBLICATIONS

---

\* indicates equal contribution. \*\* indicates alphabetical order.

### Journal articles (Published or under revision)

- [J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [pdf] [arxiv]  
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su  
*Annals of Statistics*, 53 (1): 322-351, Feb. 2025  
Selected for presentation at the Annals of Statistics Invited Paper Session, JSM 2025.
- [J2] **Robust Detection of Watermarks for Large Language Models Under Human Edits** [arxiv]  
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su  
To appear in *Journal of the Royal Statistical Society Series B*  
Received the 2025 IMS New Researcher Travel Award.
- [J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [arxiv]  
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su  
Major revision at *Journal of the American Statistical Association*.
- [J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [pdf] [arxiv]  
Yangxinyu Xie, Xiang Li, Tanwi Mallick, Weijie J. Su, Ruixun Zhang  
*Journal of the American Statistical Association*, 1–11, 2025
- [J5] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control** [arxiv]  
Xiang Li\*, Jiadong Liang\*, Xinyun Chen, Zhihua Zhang  
Minor revision at *Operation Research*

- [J6] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [\[arxiv\]](#)  
Yuze Han, **Xiang Li**, Jiadong Liang, Zhihua Zhang  
Major revision at *Mathematics of Operations Research*
- [J7] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness** [\[pdf\]](#)  
Yuze Han\*\*, **Xiang Li**\*\*, Shiyun Lin\*\*, Zhihua Zhang\*\*  
*Journal of Machine Learning Research*, 25 (380): 1-88, Dec. 2024
- [J8] **Variance-aware Decision Making with Linear Function Approximation with Heavy-tailed Rewards** [\[pdf\]](#) [\[arxiv\]](#)  
**Xiang Li**, Qiang Sun  
*Transactions on Machine Learning Research*, Apr. 2024  
Invited to present at ICLR 2025; ranked among the Top 100 papers accepted by TMLR that year.
- [J9] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation** [\[pdf\]](#) [\[arxiv\]](#)  
Xiao Guo\*, **Xiang Li**\*, Xiangyu Chang, Shusen Wang, Zhihua Zhang  
*Machine Learning*, 113 (11): 8427-8458, Sep. 2024

### Conference papers

- [C1] **On the Convergence of FedAvg on Non-IID Data** [\[pdf\]](#)  
**Xiang Li**\*, Kaixuan Huang\*, Wenhao Yang\*, Shusen Wang, Zhihua Zhang  
*International Conference on Learning Representations (ICLR) 2020* (Oral presentation, Google citation 3300+)
- [C2] **Statistical Estimation and Online Inference via Local SGD** [\[pdf\]](#)  
**Xiang Li**, Jiadong Liang, Xiangyu Chang, Zhihua Zhang  
*Conference on Learning Theory (COLT) 2022*
- [C3] **On the Empirical Power of Goodness-of-Fit Tests in Watermark Detection**  
Weiqing He\* and **Xiang Li**\*, Tianqi Shang, Li Shen, Weijie J. Su, Qi Long  
*Neural Information Processing Systems (NeurIPS) 2025* (Spotlight)
- [C4] **Mitigating the Privacy–Utility Trade-off in Decentralized Federated Learning via  $f$ -Differential Privacy**  
**Xiang Li**, Chendi Wang, Buxin Su, Qi Long, Weijie J. Su  
*Neural Information Processing Systems (NeurIPS) 2025* (Spotlight)
- [C5] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards** [\[pdf\]](#)  
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun  
*Conference on Uncertainty in Artificial Intelligence (UAI) 2025*
- [C6] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [\[pdf\]](#)  
**Xiang Li**, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan  
*International Conference on Artificial Intelligence and Statistics (AISTATS) 2023*
- [C7] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [\[pdf\]](#)  
Hengchao Chen, **Xiang Li**, Qiang Sun  
*International Conference on Artificial Intelligence and Statistics (AISTATS) 2023*
- [C8] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [\[pdf\]](#)  
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang  
*Neural Information Processing Systems (NeurIPS) 2022* (Spotlight)

- [C9] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [\[pdf\]](#)  
Shiyun Lin\*, Yuze Han\*, **Xiang Li**, Zhihua Zhang  
*Neural Information Processing Systems (NeurIPS) 2022*
- [C10] **Communication-Efficient Distributed SVD via Local Power Iterations** [\[pdf\]](#)  
**Xiang Li**, Shusen Wang, Kun Chen, Zhihua Zhang  
*International Conference on Machine Learning (ICML) 2021*
- [C11] **Finding the Near Optimal Policy via Reductive Regularization in MDPs** [\[pdf\]](#)  
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang  
*Workshop on Reinforcement Learning Theory, ICML 2021*
- [C12] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [\[pdf\]](#)  
**Xiang Li**, Shusen Wang, Zhihua Zhang  
*AAAI Conference on Artificial Intelligence (AAAI) 2020*
- [C13] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [\[pdf\]](#)  
Wenhao Yang\*, **Xiang Li\***, Zhihua Zhang  
*Neural Information Processing Systems (NeurIPS) 2019*

#### Submitted (under review)

- [S1] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [\[arxiv\]](#)  
**Xiang Li**, Jiayi Xin, Qi Long, Weijie J. Su
- [S2] **Break the Trade-off Between Watermark Strength and Speculative Sampling Efficiency for Language Models**  
Weiqing He\*, **Xiang Li\***, Li Shen, Weijie J. Su, Qi Long
- [S3] **TAB-DRW: A DFT-based Robust Watermark for Generative Tabular Data**  
Yizhou Zhao, **Xiang Li**, Peter Song, Qi Long, Weijie J. Su
- [S4] **Paraphrase-Robust Conformal Prediction for Reliable LLM Uncertainty Quantification**  
Jiayi Xin, Evan Qiang, **Xiang Li**, Weijie J. Su, Qi Long
- [S5] **High-Dimensional Nonparametric Density Estimation via Max-Random Forest**  
**Xiang Li**, Yiliang Zhang, Jiayuan Wu, Qi Long, Weijie J. Su
- [S6] **Uncertainty Quantification of Data Shapley via Statistical Inference** [\[arxiv\]](#)  
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang
- [S7] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [\[arxiv\]](#)  
Yuze Han\*\*, **Xiang Li\*\***, Zhihua Zhang\*\*
- [S8] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** [\[arxiv\]](#)  
**Xiang Li**, Jiadong Liang, Zhihua Zhang
- [S9] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [\[arxiv\]](#)  
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma

#### Working papers

- [W1] **Optimal Detection for Language Watermarks with Pseudorandom Collision**  
T. Tony Cai\*\*, **Xiang Li\*\***, Qi Long\*\*, Weijie J. Su\*\*, Garrett G. Wen\*\*

## Technical reports

- [T1] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** [\[arxiv\]](#)  
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
- [T2] **Communication-Efficient Local Decentralized SGD Methods** [\[arxiv\]](#)  
**Xiang Li**, Wenhao Yang, Shusen Wang, Zhihua Zhang

## GRANTS & FUNDING

---

- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie J. Su & Prof. Qi Long (2024)
  - Assisted in proposal writing, structuring key research points on watermarking.

## PRESENTATIONS

---

- *Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?*
  - Sep. 2025, AI Research Mixer 2025, UPenn.
- *Robust Watermark Detection and Efficient Proportion Estimation for Human-Edited Watermarked Text*
  - Jun. 2025, ICSA Applied Statistics Symposium, University of Connecticut.
  - Aug. 2025, Joint Statistical Meetings, Nashville, Tennessee.
- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
  - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
  - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
  - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
  - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
  - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.
  - Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*
  - Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.
- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
  - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*
  - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
  - Jun. 2019, the machine learning workshop at Peking University.

## ACADEMIC SERVICE

---

### Conference review

- NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.

## Journal review

- Journal of Machine Learning Research
- Transactions on Machine Learning Research
- Journal of the American Statistical Association
- Information and Inference: A Journal of the IMA
- Annals of Applied Probability
- Operational Research
- IEEE Transactions on Automatic Control
- IEEE Open Journal of Signal Processing
- IEEE Journal on Selected Areas in Communications
- Transactions on Parallel and Distributed Systems
- World Wide Web

## Event organization

- Aug. 2024, Session Chair for “Theoretical and Algorithmic Developments in Federated Learning” at the Modeling and Optimization: Theory and Applications (MOPTA) conference.

## TEACHING EXPERIENCES

---

### Instructor

- Short course (*Statistical Inference in Large Language Models*) at the 2025 ICSC Applied Statistics Symposium [\[slides\]](#) Summer 2025

### Invited guest lecturer

- STAT 9917: *Statistical Topics in Large Language Models*, UPenn Spring 2025

### Teaching assistant

- *Linear Algebra*, Peking University Spring 2021
- *Reinforcement Learning: Theory and Algorithms*, Peking University Fall 2019

## REFEREES

---

### Professor T. Tony Cai

Daniel H. Silberberg Professor  
Department of Statistics and Data Science  
The Wharton School, University of Pennsylvania  
Address: 405 Academic Research Building, 265 South  
37th Street, Philadelphia, PA 19104  
✉ [tcai@wharton.upenn.edu](mailto:tcai@wharton.upenn.edu)

### Professor Weijie J. Su

Associate Professor  
Department of Statistics and Data Science  
The Wharton School, University of Pennsylvania  
Address: 411 Academic Research Building, 265 South  
37th Street, Philadelphia, PA 19104  
✉ [suw@wharton.upenn.edu](mailto:suw@wharton.upenn.edu)

### Professor Qi Long

Professor and Vice Chair  
Department of Biostatistics, Epidemiology  
and Informatics, Perelman School of Medicine  
University of Pennsylvania  
Address: 201 Blockley Hall, 423 Guardian Drive,  
Philadelphia, PA 19104  
✉ [qlong@upenn.edu](mailto:qlong@upenn.edu)

### Professor Zhihua Zhang

Professor  
Department of Probability and Statistics  
School of Mathematical Sciences, Peking University  
Address: Science building No. 1, 5 Yiheyuan Road,  
Beijing 100871, China  
✉ [zhzhang@math.pku.edu.cn](mailto:zhzhang@math.pku.edu.cn)